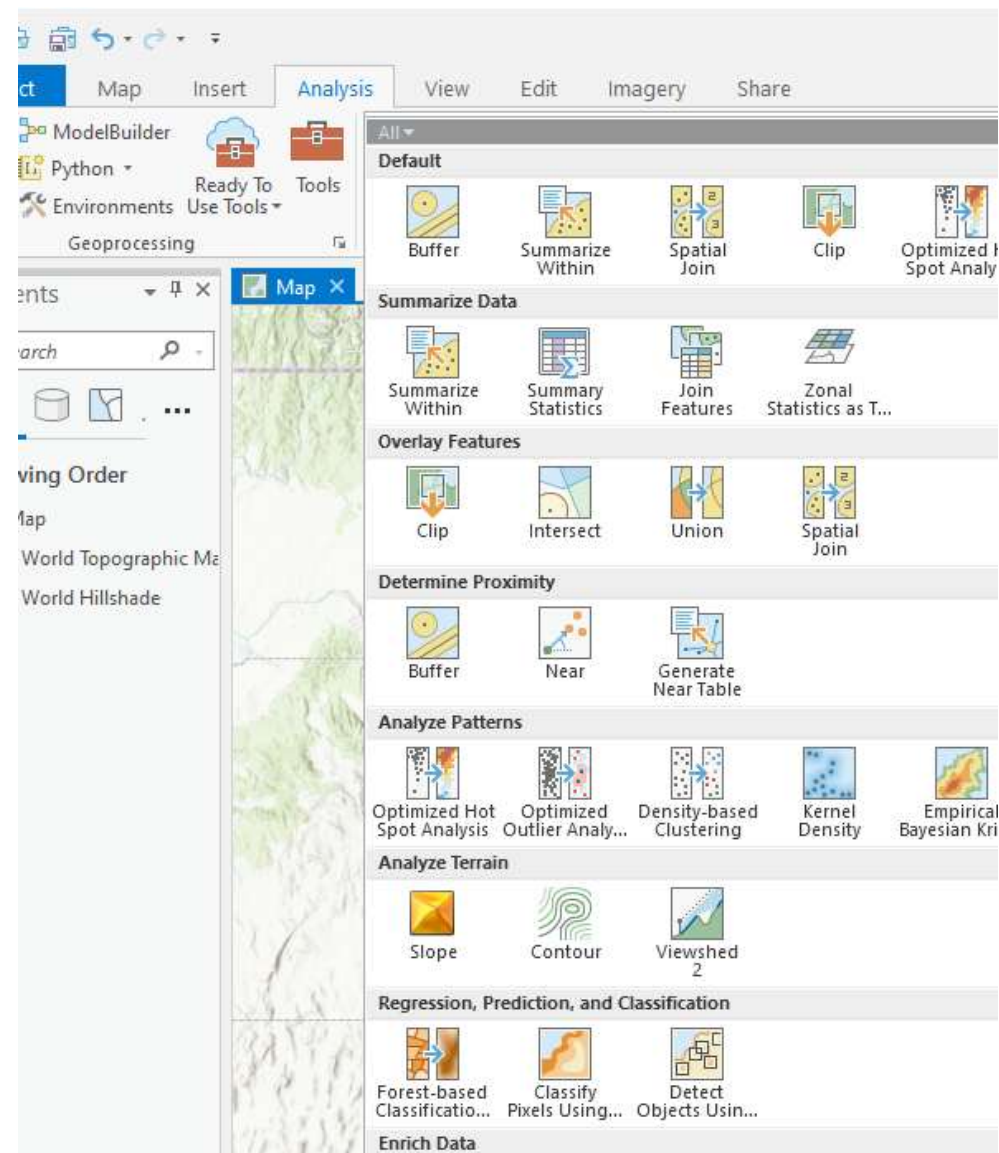


Spatial data analysis

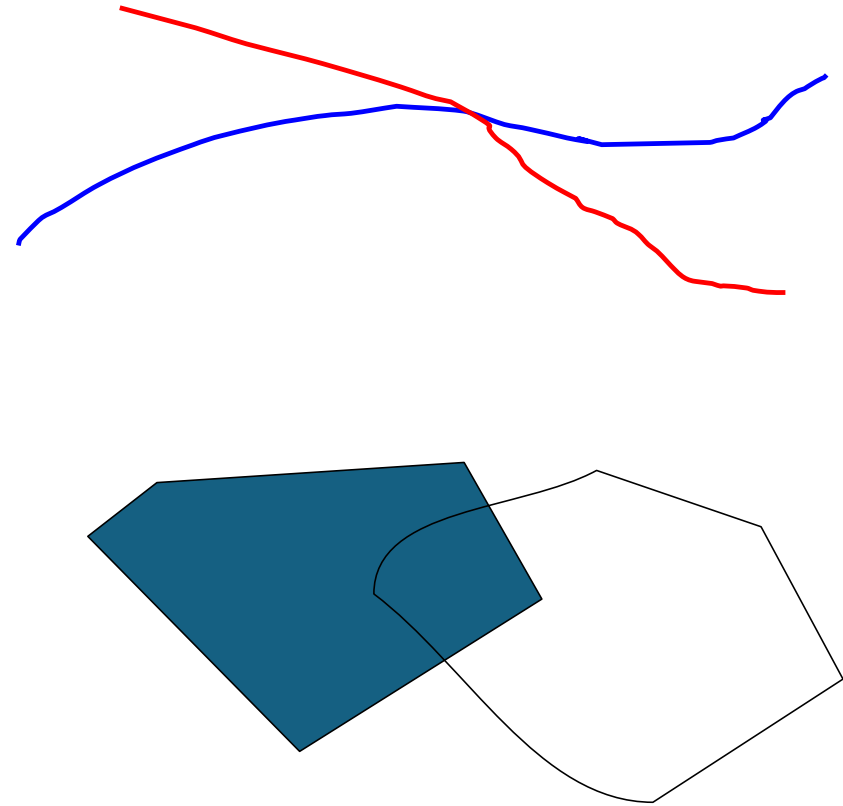
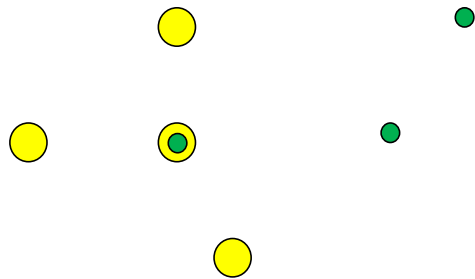
Typical operations

- **Dissolve**
- **Buffer**
- **Overlay**
- **Zonal Overlay** and Histogram
- Distance and Proximity Analysis
- Density Mapping
- Neighborhood and Block Statistics
- Interpolation and Contouring



Spatial relationships

- **Intersection**
 - Geometrically **shares a common part**
 - Unique intersection
 - Touching
 - Share a line segment with
 - Identical to (congruent)

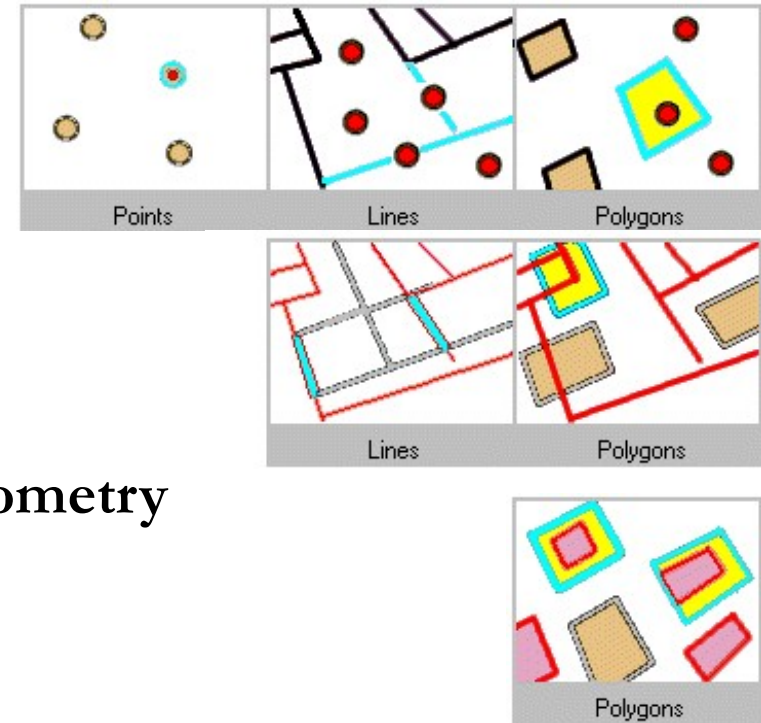


Spatial relationships

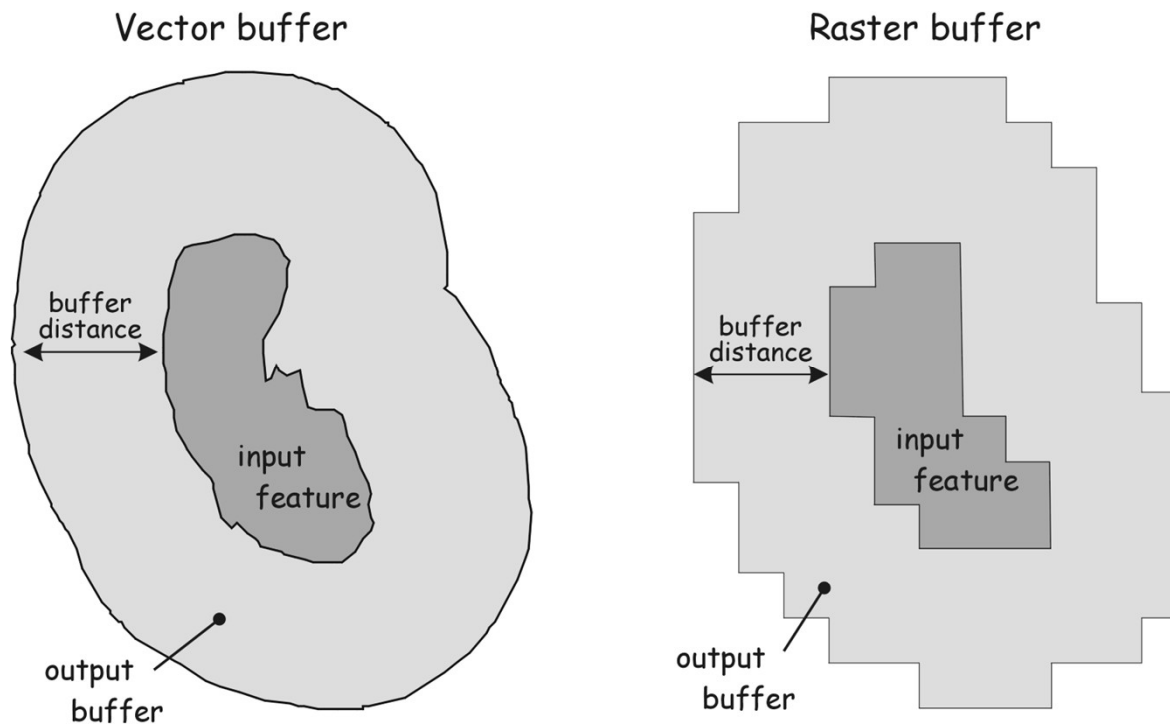
- **Containment**

- Different from Intersection:

The source feature **must fall inside the geometry** of the target feature



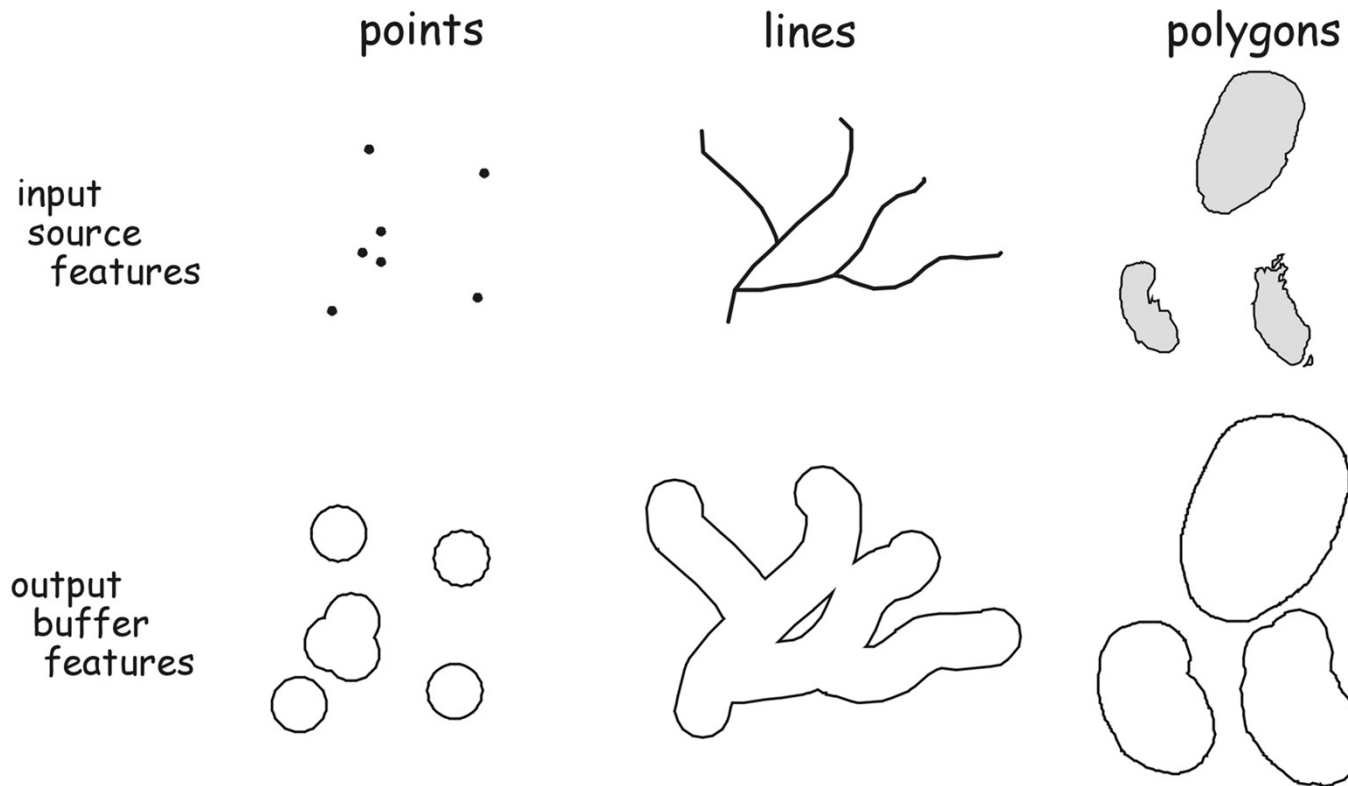
Spatial data analysis: buffer



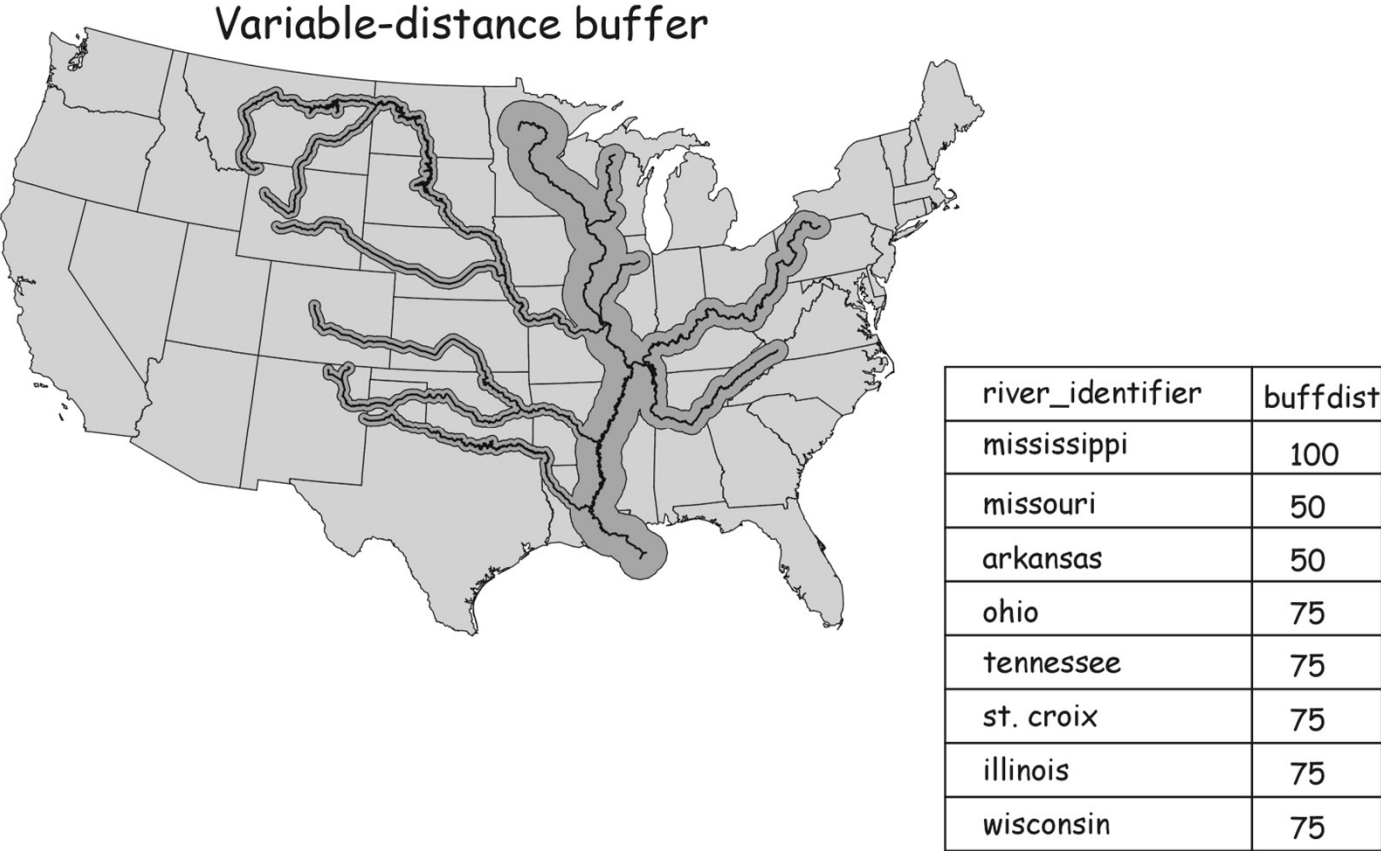
- What kinds of questions would these operations help to answer?

Spatial data analysis: buffer

Vector buffers

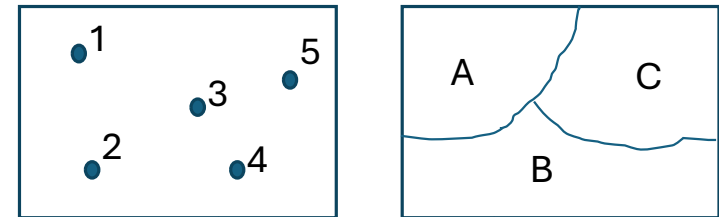
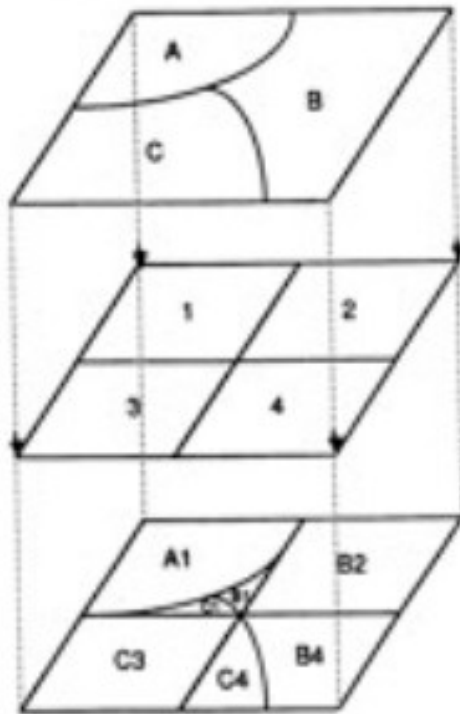


Variations on buffering



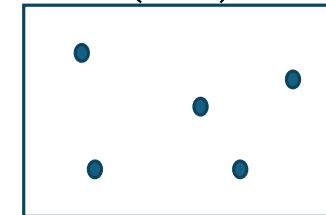
Overlay analysis

Overlay: A Closer Look



ID	City
1	Jackson
2	Rock Springs
3	Casper
4	Laramie
5	Gillette

ID	District
A	District 1
B	District 2
C	District 3

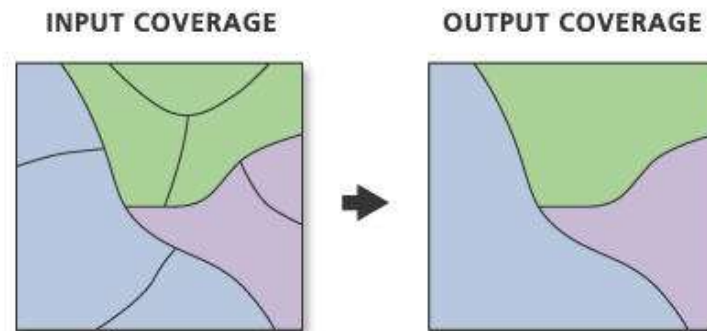


ID	City	District
1	Jackson	District 1
2	Rock Springs	District 2
3	Casper	District 3
4	Laramie	District 2
5	Gillette	District 3

Point in Polygon Overlay

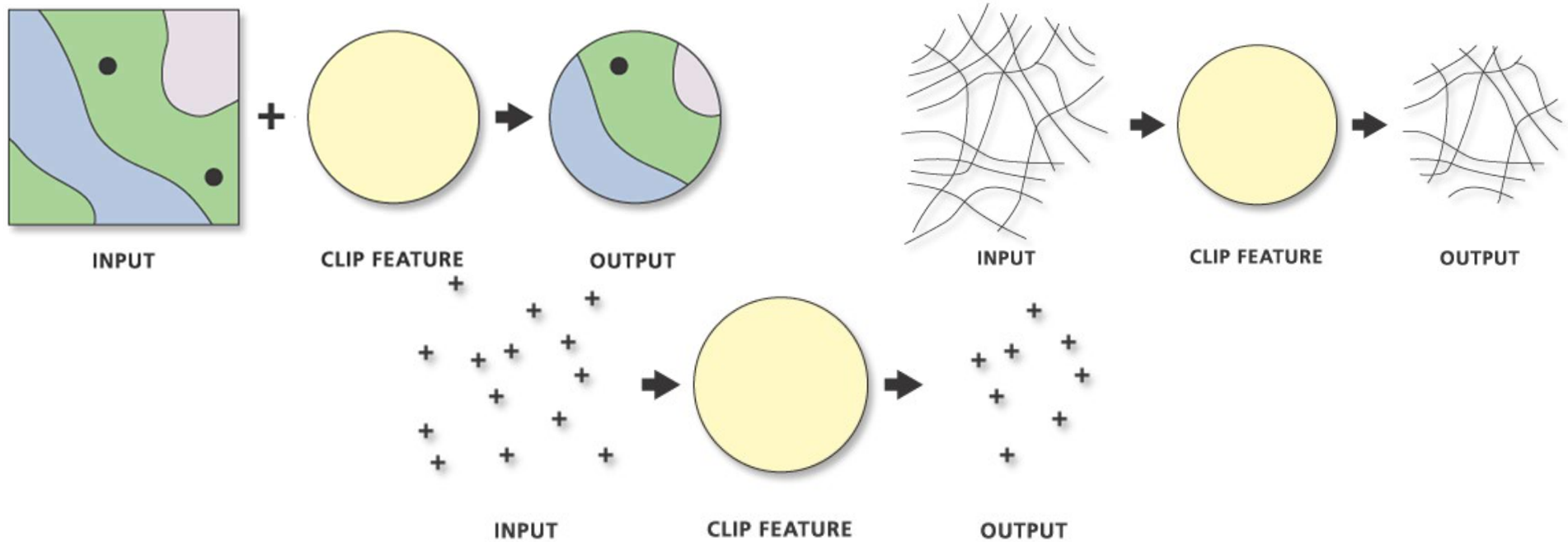
Spatial data: **dissolve**

- Combines **like** features within a data layer/feature class
- Since adjacent values of polygons may have similar attribute values, a **dissolve** removes this boundary
 - What major **advantage** does this provide?
- Use is prior to area-based selection for spatial analysis



Clip

- Think about “cookie cutters”
- **Bounding polygon** defines the **clipping layer**
 - **Spatial data** and **attributes** of clipping layer are left out of output layer



Zonal overlay

- A *zone* is all the areas/cells with the same value
- Calculate a statistic within the zones for each cell in a raster
- Input zones can be feature or raster
- Output as a raster, summary table, or graph
 - Max flow length in each watershed
 - Median income in each ZIP CODE
 - Mean elevation per vegetation zone

